could be cut if you want a shorter title - this is more a question of taste

Data in Motion: ~~Trade-Offs and Possibilities of~~ Low-Cost Motion Tracking Pipelines in Real-Time Performance

Overall this reads very well, your writing is strong! You can confidently write the paper!

Katharina Kleinhans[1]

[1]Film University Babelsberg KONRAD WOLF, Germany

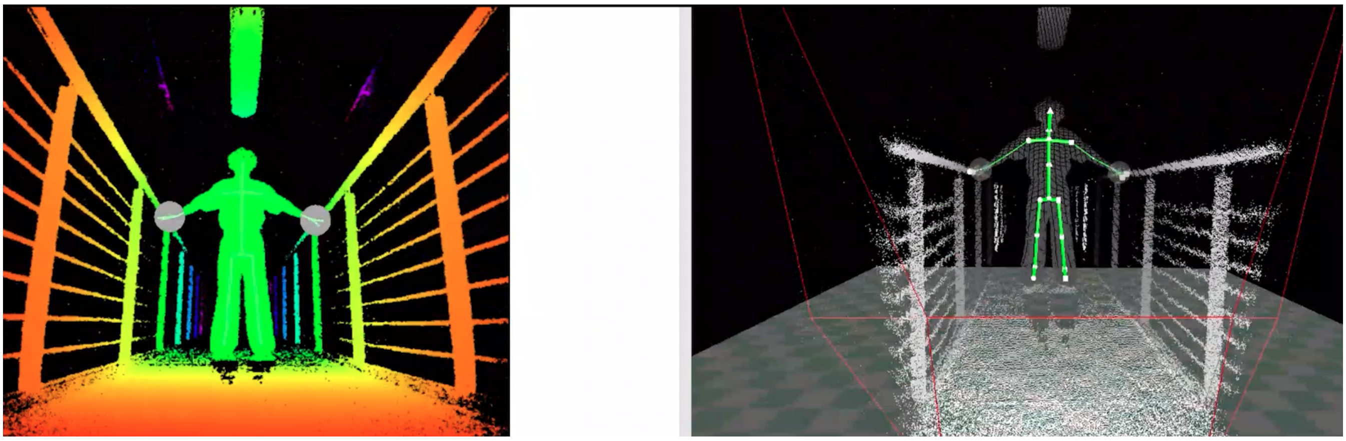

Figure 1: *Placeholder Image - Kinect Tracking*

the topic itself is open to anybody -

and with that it is

Abstract

Professional motion tracking, whether for film or live performance, typically relies on expensive, complex infrastructure ~~that~~ limits(ed) who can engage in ~~embodied interaction design~~ those devices. Affordable sensors together with open-source software, have increasingly become alternatives for artists, yet how their technical constraints shape concrete design decisions remains underexplored. This paper examines a motion-to-sound pipeline as an example developed through a collaboration between actors and creative technology students at Filmuniversität Babelsberg, combining Kinect-based skeletal tracking, relational body-data processing in TouchDesigner, and OSC/MIDI sound mapping in Ableton Live. Through iterative testing with a performer, the pipeline's design shifted from absolute to relational body-data mapping to resolve instability in tracking and gesture detection. This shift and its effect on the performer's sense of responsiveness and body awareness is examined. Results ~~in order to~~ suggest that low-cost constraints can actively shape design solutions that deepen embodied interaction, while encouraging artists to explore pipelines with affordable technology in an experimental way.

overall very good!

1. Introduction

How to go from there to "embodied interaction design"?

From Andy Serkis' performance as Gollum in *The Lord of the Rings* (1999–2003), over the Marvel universe, to the latest *Avatar* productions, Motion tracking techniques, best known for their use in film productions, have been given entirely new faces and bodies to characters. 2* This which level of precision typically relies on expensive, complex infrastructure that remains out of reach for most independent artists. However, low-budget alternatives have become increasingly popular within a growing media art culture during the

focusing on e.i.d. instead of high quality movement representation such as in blockbuster...

last years, as open-source softwares and affordable sensors offer accessible entry points into embodied interaction design. Depth cameras built into gaming consoles, such as Microsoft's Xbox Kinect, have entered our everyday lives and have become a breeding ground for experimental exploration of the body's own movement - a ~~new~~ technical bridge between digital and performative arts.

good!

Like any form of art, this intersection of the arts and technology requires certain technical skills: just as a visual artist needs a brush and a surface to paint on, and a musician plays an instrument, me-

or hardware? The following examples list "devices" not skills...

* you mean "the movement for..."

Do not repeat text between abstract and in

dia artists need various hardware and software to bring their ideas to life. Specifically, working with the movement of bodies requires stable solutions for capturing movement as well as processing the data. Establishing a system for comprehensive body tracking and reliable data processing is a challenging task in itself. Why? Unlike art forms that rely on physical materials, these tools are often more complex because and influenced by additional factors: the state of the art in development, their accessibility and availability, and the shared knowledge surrounding their use. Factors, that can rise the barrier for artists to work with these mediums. While professional motion-tracking systems have been perfected over the years, working with low budget pipelines still comes with various obstracles, as will be further analyzed by the case study in this paper.

I don't think this comparison holds. Just write this description not comparative

to what?

Several studies have demonstrated that affordable alternatives to professional motion-tracking setups can lead to meaningful results. Open-source pipelines have been used to design inclusive systems that lower entry barriers for disabled practitioners [Cha23], while Kinect-based systems have been applied in clinical rehabilitation to support and track patient recovery [CSH*17]. These examples suggest that budget-conscious motion capture technology does not have to be a compromise, but can enable practical applications precisely because of its accessibility. For media artists, this accessibility opens up similar possibilities: low-cost sensors allow artists to extract the techniques from its original professional context and place it within new artistic contexts, from multimedia installations to interactive applications and live performance, particularly within a culture of low-budget creative practice. Yet how these constraints concretely shape the artistic process, like trade-offs, workarounds, and design decisions involved, remains underexplored.

As a case study...

This paper examines a motion-to-sound pipeline developed through a collaboration between actors and creative technology students at Filmuniversity Babelsberg.* By documenting the technical challenges and design solutions encountered, the paper reveals how low-cost motion-tracking constraints can shape creative and embodied possibilities in real-time performance.

* why is this a suitable an representable case?

2. Related Work

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3. Main Content - Name and Structure the Main Content According to your Paper

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3.1. Method

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3.2. Results

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3.3. Discussion

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4. Formatting your paper

4.1. Bullet Points

The following is an example for a bullet points list. The contributions of this paper are

- my greate archivement number one,
- a second awsome insight,
- and last but not least the third contribution.

4.2. Figures

All graphics should be centered.

Figure 2: *Here is a sample figure. All figures must have sensible captions and must be referenced in the text.*

If your paper includes images, it is important that they are of sufficient resolution to be faithfully reproduced.

Instructions from the original conference template (for the ACS FUB it is not that relevant):

To determine the optimum size (width and height) of an image, measure the image's size as it appears in your document (in millimeters), and then multiply those two values by 12. The resulting values are the optimum x and y resolution, in pixels, of the image. Image quality will suffer if these guidelines are not followed.

Example 1: An image measures 50 mm by 75 mm when placed in a document. This image should have a resolution of no less than 600 pixels by 900 pixels in order to be reproduced faithfully.

This is an examplary reference to Figure 3, which shows the incredible test image from the Eurographics conference.

4.3. Tables

A simple table might look like the following. However there are several ways to create and layout tables - it is up to you to style a table up to your liking. This is a reference to Table 1.

Col1	Col2	Col2	Col3
1	6	87837	787
2	7	78	5415
3	545	778	7507
4	545	18744	7560
5	88	788	6344

Table 1: *Here is a sample table. All tables must have sensible captions and must be referenced in the text.*

4.4. Formulas

In LaTeX it is fairly easy to add mathematical symbols and formulas. See the following as example [**?**]:

$$a = bq + r \tag{1}$$

which is true if a and b are integers with $b \neq c$.

$$L' = L\sqrt{1 - \frac{v^2}{c^2}} \tag{2}$$

4.5. Footnotes

Please do not use footnotes at all.

4.6. References

References are layouted automatically. Add the needed entry to the bibtex file and use the citation command to reference the entry.

If you don't want to mention the author's name just add the citation to the end of the sentence [**?**]. You can also cite more than one references at the same time [**?**, **?**, **?**].

Sometimes you want to refer to the authors name. With this template you need to write out the names manually. For more than two authors you only use the name of the first author and abbrivate the remaining names with "et al."

An example for more than two authors:

Raul et al. [**?**] thoroughly review the different approaches for segmentation algorithms. Our problem requires the segmentation of color images, summarized by Cheng et al. [**?**].

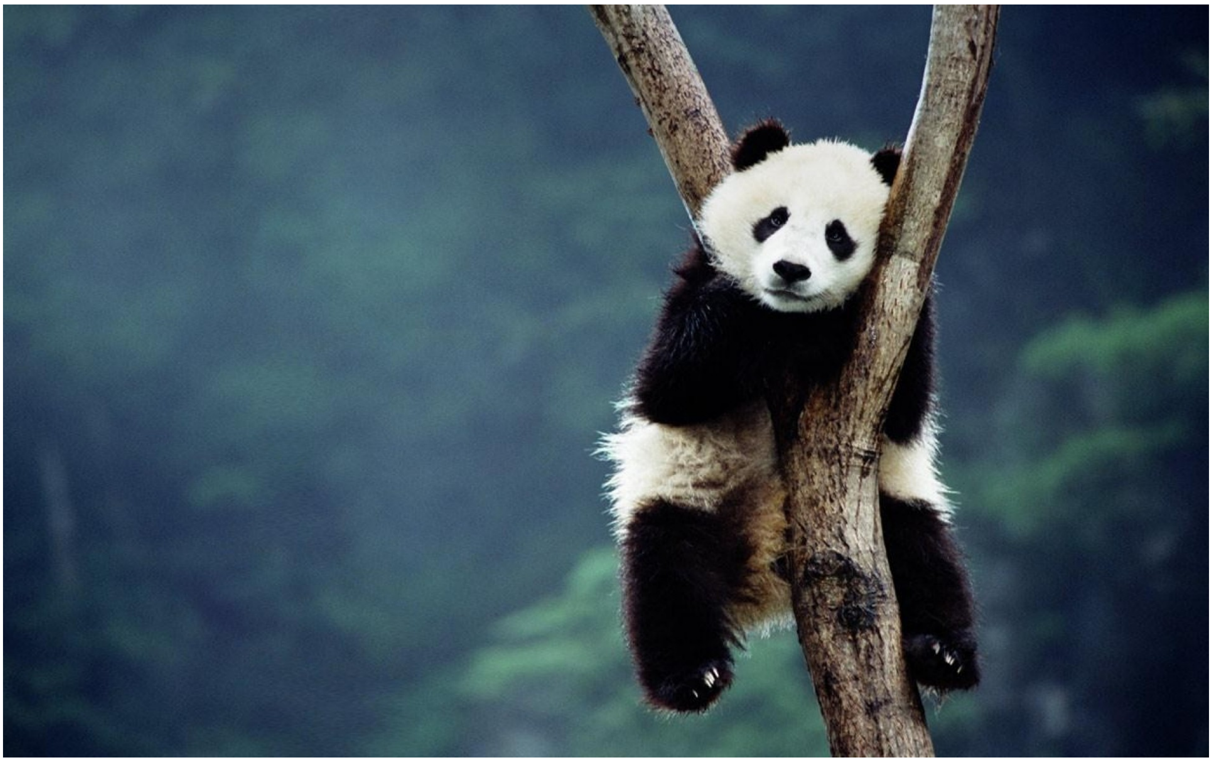

Figure 3: *Two-column width figure*

An example for two authors:

Also aiming for a fragmented aesthetic appeal by transforming original image data "piecewise", Collomosse and Hall [**?**] render from salient image features a cubist version of an input. Similarly, Lai and Rosin [**?**] loosely aim for a cubist appearance by segmenting an input and by simplifying the segments into constant colors, while maintaining global structures.

An example for one author:

Artistically, we are inspired by the art of Wehrli [**?**], who manually creates image decompositions and rearrangements.

5. Conclusion

Please direct any questions to the production editor in charge of these proceedings.

References

[Cha23] CHANG C.: *Enhancing the Accessibility and Usability of Motion Capture Technology: Design and Development of Indoor MoCap Hardware System*. Master's thesis, Massachusetts Institute of Technology, 2023. 2

[CSH*17] CHANPIMOL S., SEAMON B., HERNANDEZ H., HARRIS-LOVE M., BLACKMAN M. R.: Using xbox kinect motion capture technology to improve clinical rehabilitation outcomes for balance and cardiovascular health in an individual with chronic tbi. *Archives of Physiotherapy 7*, 1 (2017), 6. 2